

SIMATS ENGINEERING

Assessment -CO2-AT1

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP,*

Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Subject	Artificial Intelligence
Course Code	CSA 1712
Name	Vignesh Kumar K
Register Number	192512375

Declaration

I, Vignesh Kumar K (Reg No: 192512375), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

MCQ Answer Key — All 20 Questions

The correct option is shown in bold and highlighted; a brief one-line rationale follows each question.

Q1. In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next, focusing on the node that appears closest to the goal without considering the cost already incurred. Which algorithm follows this principle?

- A) Breadth First Search
- B) Uniform Cost Search

C) Greedy Best-First Search ✓ Correct Answer

- D) Depth First Search

Rationale: Greedy Best-First Search expands purely by $h(n)$, the estimated distance to the goal, ignoring path cost so far.

Q2. A* search combines the actual cost from the start node and the estimated cost to the goal, allowing it to find optimal solutions efficiently when the heuristic is admissible. What is the evaluation function used in A* search?

- A) $f(n) = g(n)$
- B) $f(n) = h(n)$

C) $f(n) = g(n) + h(n)$ ✓ Correct Answer

- D) $f(n) = g(n) \times h(n)$

Rationale: A* evaluates nodes using $f(n) = g(n) + h(n)$ — actual cost so far plus estimated remaining cost.

Q3. A good heuristic should not overestimate the actual cost and should improve search efficiency. What property ensures optimality in A* search?

- A) Completeness

B) Admissibility ✓ Correct Answer

- C) Depth limit

- D) Backtracking

Rationale: An admissible heuristic (never overestimates true cost) guarantees A* finds the optimal solution.

Q4. Local search algorithms start with an initial solution and iteratively improve it based on an objective function, commonly used where the path is not important but the final solution is. Which of the following is a local search algorithm?

- A) A* Search

B) Hill Climbing ✓ Correct Answer

- C) Breadth First Search

- D) Uniform Cost Search

Rationale: Hill Climbing iteratively moves to better neighbouring states based on an objective function — a classic local search method.

Q5. In adversarial search, agents compete in a game environment, considering both their own moves and the opponent's possible responses. Which algorithm is commonly used for such decision-making?

- A) Greedy Search

B) Minimax Algorithm ✓ Correct Answer

- C) Backtracking

- D) Uniform Search

Rationale: Minimax models the adversarial game tree, assuming the opponent plays optimally to minimize the agent's outcome.

Q6. A delivery robot in a warehouse selects paths based only on the estimated distance to the destination without considering obstacles already encountered, sometimes leading to suboptimal but faster decisions. Which algorithm is the robot using?

A) A* Search

B) Greedy Best-First Search ✓ Correct Answer

C) Uniform Cost Search

D) Depth Limited Search

Rationale: Choosing paths purely by estimated remaining distance, ignoring cost already incurred, is the defining behaviour of Greedy Best-First Search.

Q7. A GPS navigation system calculates routes by combining the distance already travelled and an estimate of the remaining distance to the destination to find the shortest path efficiently. Which algorithm best represents this approach?

A) Depth First Search

B) A* Search ✓ Correct Answer

C) Hill Climbing

D) Backtracking

Rationale: Combining cost-so-far with an estimate of remaining cost is exactly A's $f(n) = g(n) + h(n)$ approach.*

Q8. A university is scheduling exams for multiple subjects with constraints such as no overlapping exams and limited rooms, and the solution must satisfy all constraints while assigning time slots. This problem is best modeled as:

A) Game Playing Problem

B) Constraint Satisfaction Problem ✓ Correct Answer

C) Heuristic Search Problem

D) Optimization Problem

Rationale: Variables (exams) with domains (timeslots/rooms) and constraints (no overlap, room limits) is the definition of a CSP.

Q9. An AI-based chess program evaluates all possible moves and anticipates the opponent's responses before making a decision, assigning utility values to game states to determine the best move. Which algorithm is being applied here?

A) Greedy Search

B) Hill Climbing

C) Minimax Algorithm ✓ Correct Answer

D) Breadth First Search

Rationale: Evaluating game states with utility values while anticipating an adversary's optimal responses is the Minimax approach.

Q10. A robot explores an unknown environment where it does not have prior knowledge of obstacles, making decisions step-by-step and updating its knowledge as it moves. Which type of agent is suitable for this scenario?

A) Offline Agent

B) Online Search Agent ✓ Correct Answer

C) Static Agent

D) Deterministic Agent

Rationale: An online search agent interleaves sensing, planning, and acting, building its knowledge of the environment as it explores — exactly this scenario.

Q11. A* search guarantees optimal solutions only when certain conditions are satisfied by the heuristic function. Why is an admissible heuristic important?

A) It reduces memory usage

B) It ensures optimality ✓ Correct Answer

C) It increases branching factor

D) It avoids loops

Rationale: Admissibility (never overestimating true cost) is precisely the condition that guarantees A's optimality.*

Q12. In large game trees, evaluating every node becomes computationally expensive. Alpha-Beta pruning helps eliminate branches that do not affect the final decision. What is the primary benefit of Alpha-Beta pruning?

A) Reduces depth of tree

B) Reduces number of nodes evaluated ✓ Correct Answer

C) Changes game rules

D) Improves heuristic accuracy

Rationale: Alpha-Beta pruning skips branches that cannot influence the final decision, reducing the number of nodes evaluated without changing the result.

Q13. Hill Climbing algorithm continuously moves toward better states based on local improvements, but it may fail to reach the global optimum due to certain limitations. What is the main drawback of Hill Climbing?

A) High memory usage

B) Revisiting nodes

C) Getting stuck in local maxima ✓ Correct Answer

D) Requires complete tree

Rationale: Hill Climbing has no mechanism to escape a state where all neighbours are worse, even if a better global solution exists elsewhere — it gets stuck at local maxima.

Q14. Constraint Satisfaction Problems involve variables, domains, and constraints, and a valid solution must assign values to all variables such that no constraints are violated. What ensures a valid CSP solution?

A) Heuristic function

B) Path cost

C) Constraint satisfaction ✓ Correct Answer

D) Depth-first traversal

Rationale: By definition, a CSP solution is valid precisely when every constraint over the assigned variables is satisfied.

Q15. In adversarial games, evaluation functions are used when it is not feasible to explore the entire game tree, estimating the desirability of a game state. What is the role of an evaluation function?

A) Generate successors

B) Estimate utility of a state ✓ Correct Answer

C) Reduce branching factor

D) Store moves

Rationale: An evaluation function assigns an estimated utility/desirability score to a non-terminal game state when full search isn't feasible.

Q16. In A* search, the evaluation function is calculated using both the path cost and heuristic value. Suppose a node has a path cost of 7 and a heuristic estimate of 5. What is the value of $f(n)$?

A) 12 ✓ Correct Answer

B) 2

C) 35

D) 7

Rationale: $f(n) = g(n) + h(n) = 7 + 5 = 12$.

Q17. A heuristic function sometimes overestimates the actual cost to reach the goal, violating the admissibility condition required for optimal solutions in A* search. What happens in such a case?

A) A* remains optimal

B) A* becomes non-optimal ✓ Correct Answer

C) A* becomes BFS

D) A* stops working

Rationale: Without admissibility, A* can no longer guarantee it finds the optimal-cost solution — it may return a suboptimal path.

Q18. Consider a search tree with a branching factor of 2 and depth of 4. What is the approximate number of nodes?

A) 10

B) 15

C) 31 ✓ Correct Answer

D) 16

Rationale: Full tree node count = $\sum_{i=0}^4 2^i = 1+2+4+8+16 = 31$.

Q19. In a minimax tree, each node has 3 children and the depth of the tree is 3. How many leaf nodes are present?

A) 9

B) 27 ✓ Correct Answer

C) 6

D) 12

Rationale: Leaf nodes = $\text{branching factor}^{\text{depth}} = 3^3 = 27$.

Q20. Alpha-Beta pruning improves the efficiency of minimax by reducing the number of nodes evaluated. What is the best-case time complexity?

A) $O(b^d)$

B) $O(b^{(d/2)})$ ✓ Correct Answer

C) $O(d^b)$

D) $O(\log b)$

Rationale: With optimal move ordering, Alpha-Beta pruning achieves a best-case time complexity of $O(b^{(d/2)})$, effectively doubling feasible search depth.